



CO hydrogenation on group VI metal–ceria catalysts

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ABSTRACT

Ceria-supported chromium, molybdenum, and tungsten catalysts were prepared by impregnation. The prepared catalysts were characterized using N₂ adsorption, X-ray diffraction (XRD), the temperature-programmed reduction (H₂-TPR), and X-ray photoelectron spectroscopy (XPS) measurements. The catalytic activity in CO hydrogenation was evaluated using a fixed-bed pressurized flow reaction system under the following conditions: 260–300 °C, 5.0 MPa, GHSV of 5000 h^{−1}, and a H₂/CO ratio of 1.0–2.0. The effects of ceria support, group VI metals, and catalyst activation methods on C₂₊ alcohol synthesis were investigated. The use of ceria supports resulted in a decrease in the selectivity for CO₂, and in increases in the selectivity for C₂₊ alcohols and CO conversion. The selectivity for alcohols on the Mo-based catalysts was higher than those on the corresponding Cr or W-based catalysts. A comparison of the methods of activation for the K_{0.05}Co_{0.6}MoCe catalyst demonstrated that sulfidation produced the highest CO conversion and selectivity for C₂₊ alcohols, as well as the lowest, hydrocarbon selectivity. XPS and H₂-TPR measurements show that the mixed metal sulfide phases, e.g., the Co–Mo–S phase, and the thiol group on the catalysts enhanced the formation of C₂₊ alcohols.

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1. Introduction

Alcohols with more than two carbons, described as C₂₊ alcohols or mixed alcohols, are extensively utilized as chemical intermediates, such as plastics, solvents, paint, and aromatics, as well as alternative fuel. The demands for C₂₊ alcohols are increasing year by year. Currently, C₂₊ alcohols are synthesized by hydration or hydroformylation of olefins, which are derived from fossil resources [1]. Recently, it has become necessary to manufacture C₂₊ alcohols from sources other than fossil resources. Because the biomass wastes or residues can be used as feedstock for the production of C₂₊ alcohols via gasification to syngas, which is a mixture of hydrogen and carbon monoxide, the manufacture of C₂₊ alcohols does not need to rely solely on fossil resources. Therefore, considerable attention has been focused on the process of transforming syngas to mixed alcohols [2,3].

Noble metals, such as rhodium, exhibit good catalyst performance with their selectivity for C₂₊ alcohols [4,5]. However, these catalysts are sensitive to sulfur poisoning [6]. Thus, from a cost perspective, it is essential to explore non-noble metal catalysts for alcohol synthesis applications. Molybdenum-based catalysts are considered a promising candidate for the synthesis of mixed alcohols [2,3,7–9]. It is well known that the addition of transition metals, such as cobalt, as promoters to sulfided Mo/Al₂O₃ or MoS₂ drastically improves hydrodesulfurization (HDS) activity, and that WS₂ or Cr₂S₃-based catalysts have similar activity to those of MoS₂-based catalysts [10–13]. The addition of a transition metal to catalysts based on group

VI metals also has the potential to promote the synthesis of mixed alcohols, e.g., by improving the propagation of carbon chains. This potential comes from the fact that transition metals, such as Fe, Co, and Ni, are also used as catalysts for F–T synthesis. The addition of group VIII metals, especially cobalt, on unsupported alkali–MoS₂ catalysts was reported to shift the selectivity for methanol toward C₂₊ alcohols and to decrease the selectivity for hydrocarbons [14]. Similar effects of adding cobalt were also shown in our previous study [7]. Therefore, alkali and Co-doped catalysts based on group VI metals could be expected to produce more C₂₊ alcohols.

In our previous study, we investigated the effects of acidity on catalysts during the synthesis of mixed alcohols [7]. Lower amounts of acid and an appropriate loading of K on the catalysts promoted the formation of C₂₊ alcohols. Ceria, which is a basic oxide, is available as a catalyst support for the exhaust-gas treatment of diesel engines and the steam reforming reaction, and it is known as a reducing agent [15,16]. It was also reported that ceria supports possessed reducibility and a redox equilibrium with Cu (Ce⁴⁺ + Cu⁰ ⇌ Ce³⁺ + Cu⁺) in the synthesis of C₂₊ alcohols over CsCuCeO₂ catalysts [17]. Because MoS₂-based catalysts possess the relatively high activity necessary for water–gas-shift reactions, they can readily form CO₂ [3,9]. It was proposed that CO₂ is formed via the surface catalytic reaction of the non-dissociatively adsorbed CO species, or CO molecule, with O* species derived from dissociatively adsorbed C–O species [3,7,18]. Therefore, it is likely that a ceria support could accept the dissociated oxygen species, and the formation of CO₂ might be suppressed on ceria-supported catalysts.

The influence of group VI metals on catalysts with a ceria support has not been elucidated with regard to the synthesis of C₂₊ alcohols. Thus, to elucidate those effects in the present study, we focused on

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evaluating the abilities of various K–Co-group-VI–metal–ceria catalysts to synthesize C_2+ alcohols. A series of catalysts with ceria supports were prepared by impregnation, and the roles of ceria and the catalyst activation methods were investigated.

2. Experimental

2.1. Materials

Ceria in power form (Daiichi Kigenso Kagaku Kogyo Co., Japan) and SiO_2 (Japan Oil Co.) were purchased and used as supports. For preparation, commercial GR grade hexaammonium heptamolybdate 4-hydrate ($(NH_4)_6Mo_7O_{24} \cdot 4H_2O$), chromium nitrate nonahydrate ($Cr(NO_3)_3 \cdot 9H_2O$), cobalt nitrate hexahydrate ($Co(NO_3)_2 \cdot 6H_2O$), and potassium precursors: potassium nitrate (KNO_3) or potassium carbonate (K_2CO_3) (Wako Pure Chemicals, Japan, respectively), were used, and commercial 1 grade ammonium tungstate para pentahydrate ($(NH_4)_{10}W_{12}O_{41} \cdot 5H_2O$ (88.0%)) was used. The feed gas was a mixture of a standard mix of $H_2/CO/N_2$ (45/45/10 vol.%) and hydrogen (purity: 99.999 vol.%).

2.2. Preparation of catalysts

All catalysts were prepared by the conventional and sequential impregnation procedure according to our previous study [7]. Ceria or SiO_2 support was impregnated with aqueous solution of $(NH_4)_6Mo_7O_{24} \cdot 4H_2O$, $Cr(NO_3)_3 \cdot 9H_2O$, or $(NH_4)_{10}W_{12}O_{41} \cdot 5H_2O$ as group VI metal precursors firstly, and was evaporated to dryness. Next, the aqueous solution of $Co(NO_3)_2 \cdot 6H_2O$ was introduced successively. The impregnated samples were pressurized for shaping under 60 MPa for 5 min and then pulverized to 20–80 mesh. After calcining in air at 450 °C for 12 h, the Co-group VI metal oxide precursor was impregnated with a solution of K_2CO_3 or KNO_3 , being evaporated to dryness, and calcined in air at 450 °C for 12 h in the same way. The loading amounts of K_2O (or K_2CO_3), CoO, and group VI metal oxides are listed in Table 1. For investigation of the effects of group VI metals, the ratio of group VI metal to ceria support was constant at 0.23 (mol/mol). The prepared samples were denoted as $K_xCo_yzM_Ce$ (x : molar ratio of K/M; y : molar ratio of Co/M; z : weight of group VI metal (wt.%); M : group VI metal (Cr, Mo, or W), abbreviation to parenthetic C in the case of K_2CO_3 , and only “K” in the case of KNO_3). All catalysts were pretreated according to one of three procedures. In the sulfidation procedure, all samples were sulfided in situ with a mixed gas of 5 vol.% H_2S and 95 vol.% H_2 at a flow rate of 30 ml/min at 400 °C at temperature ramp rate of 5 °C/min and atmospheric pressure for 3 h. For reduction of the $K_{005}Co_{06}20MoCe$ sample, hydrogen (purity of 99.999 vol.%) was substituted for a mixed gas of H_2S/H_2 at the same flow rate, pretreatment temperature and temperature rate, and time as those used for sulfidation. The $K_{005}Co_{06}20MoCe$

sample was also pre-sulfided at 400 °C for 3 h and was then reduced at 320 °C for 3 h to investigate the effects of the catalyst activation methods.

2.3. Characterization of catalysts

The specific surface area and pore size of all samples were determined using N_2 adsorption with an automatic specific surface area and pore volume measurement device (BELSORP-mini II, BEL Japan) [19]. X-ray diffraction (XRD) measurements were performed using an X-ray diffractometer (RAD-IIC, Rigaku Co.) with Cu $K\alpha$ radiation. A continuous scan mode was used to collect 2θ data from 5° to 80° at the rate of 1.5°/min. The voltage and current were 40 kV and 30 mA, respectively. The reducibility of the sulfided catalysts was measured with the temperature-programmed reduction (H_2 -TPR) using a ChemBET Pulsar TPR/TPD instrument (Quantachrome Instruments). Before H_2 -TPR measurements, several samples (0.1 g) were pretreated as described above in Section 2.2. H_2 -TPR analyses were performed by a previously published method [7]. H_2S molecules, which were adsorbed from sulfided samples, were trapped with grained Fe_2O_3 as adsorbent. Prior to X-ray photoelectron spectroscopy (XPS) analyses, all samples were pretreated in a quartz reactor. The XPS measurements of the catalysts were taken using a Shimadzu ESCA 3200 photoelectron spectrometer equipped with a magnesium source ($Mg K\alpha = 1253.6$ eV) that was operated at 8 kV and 30 mA under previously published conditions [20]. The envelopes of Mo 3d, Co 2p, S 2p, K 2p, O 1s, C 1s, and Ce 3d were measured and those spectra were deconvoluted with a curve-fitting based on a Gaussian function.

2.4. CO hydrogenation and analysis

Prior to activity tests, all catalysts were pretreated as described above in Section 2.2. The tests for alcohol synthesis from syngas were performed using a fixed-bed pressurized flow reaction system with a micro-reactor of conventional design. The detailed experimental procedures and sampling methods followed those of a previously published source [7]. The reaction conditions were 260–300 °C, 5.0 MPa, gas hourly space velocity (GHSV) of 5000 h^{-1} , and H_2/CO ratio of 1.0 or 2.0. An online gas chromatograph was equipped with FID (Model GC-14B, Shimadzu; CP-Sill 5CB, 0.25 mm $\times$ 60 m length) to analyze any oxygenated compounds produced, such as alcohols and hydrocarbons. The oxygenated compounds as esters were not detected regarding all samples under the detectable analysis condition. Another online GC-TCD (Unibeads C, 1/8 in. $\times$ 2.0 m length, GC-323, GL Science Inc.) was equipped to analyze CO as the reactant and the CO_2 and C_{1-4} light hydrocarbons that were produced. The catalytic performance was also evaluated by CO conversion (%), the selectivity based on carbon for a product (%-C based), space-time-yield (the yield per a weight of

Table 1
Physical properties of the prepared catalysts and ceria supports.

Sample	K_2O or K_2CO_3 (wt.%)	CoO (wt.%)	Group VI metal oxides ^a (wt.%)	S_{BET} ^b (m^2/g)	V_p ^c (cm^3/g)	d_a ^d (nm)	d_{peak} ^e (nm)
Ceria	–	–	–	139	0.19	5.4	3.3
$K_{02}Co_{06}10MoSi$	0.65	3.1	10	177	0.84	19	24
$K_{02}Co_{06}10MoCe$	0.65	3.1	10	71.2	0.14	8	5.4
$K_{005}Co_{06}11MoCe$	0.16	3.1	10	71.6	0.14	8	5.4
$K_{005}Co_{06}15MoCe$	0.25	4.7	15	63.6	0.21	13.1	8.1
$K_{005}Co_{06}20MoCe$	0.33	6.3	20	48.3	0.13	11	9.2
$K(C)_{005}Co_{06}8.6CrCe$	0.39	5.1	8.6	91.1	0.17	7.6	5.4
$K(C)_{005}Co_{06}15MoCe$	0.36	4.7	15	63.9	0.16	9.8	7.1
$K(C)_{005}Co_{06}22WCo$	0.33	4.3	22	58.9	0.15	10	4.8

^a Group VI metal oxides: MoO_3 , WO_3 , or Cr_2O_3 .

^b S_{BET} : BET surface area.

^c V_p : pore volume.

^d d_a : average pore diameter.

^e d_{peak} : maximum pore diameter.

catalyst [mol/h/kg-cat.]) of alcohols, and the probability of carbon chain growth in hydrocarbons and alcohols [7].

3. Results

3.1. Characterization of catalysts

The textural properties of the prepared catalysts and ceria support are listed in Table 1. The surface area of $K_{0.05}Co_{0.6}10MoSi$ was larger than that of $K_{0.05}Co_{0.6}10MoCe$. As the Mo loading amount increased, the surface area decreased from 71.6 to 48.3 m²/g. The decline in surface area, when the loading amount was increased from 15 to 20%, was larger than that which increased from 10 to 15%. The addition of a large amount of metal led to a decrease in specific surface area, indicating that the dispersion of metal for the 20% Mo loading might be lower than that for Mo loads of less than 15% on the corresponding catalysts.

XRD patterns of the oxidized, reduced, and sulfided catalysts and the ceria supports were measured and are shown in Fig. 1. In each sample, several peaks occurred with strong intensities on all catalysts and were similar to those of the ceria supports. The peaks at 21.9°, 25.0°, 30.0°, 40.0°, 45.0°, 47.8°, 54.3°, and 57.0° appeared in all sample patterns. No other peaks corresponding to the lattice of active metals were observed. These results indicate the high dispersion of metals on all samples.

To investigate the effects of Mo loadings, the supports, and the activation methods, H₂-TPR profiles were performed and are shown in Fig. 2. A dominant peak appeared when the Mo loading was changed from 10% to 20%, as shown in Fig. 2A (a)–(c). The peak at the range of 530–634 °C was ascribed to the partial reduction of Mo⁶⁺ to Mo⁴⁺ of amorphous, highly defective, multilayered Mo oxides or heteropolymolybdates and octahedral Mo species [21–23]. With increasing the Mo loading, the main peak shifted to the high temperature from 530 to 634 °C, and the peak area increased, indicating an increase in H₂ consumption and the formation of metal species, i.e. bulk MoO₃, which was difficult to reduce on the catalysts.

The profiles for the different supports are given in Fig. 2A (d)–(f). Two peaks appeared: one at low temperatures ranging from 509–620 °C and another at the high temperature of 864 °C. The low-temperature peak was assigned to the partial reduction of Mo⁶⁺ to Mo⁴⁺ on the catalysts. The peak at 864 °C was generally related to the deep reduction of Mo⁴⁺ to Mo, including the highly dispersed tetrahedral Mo species strongly bound to the support [24,25]. The high-temperature peak at 864 °C for the ceria-supported sample was smaller than that for the SiO₂ or Al₂O₃ supported samples. This finding means that the ceria support had weak interaction with the metals compared to the SiO₂ and Al₂O₃ supports.

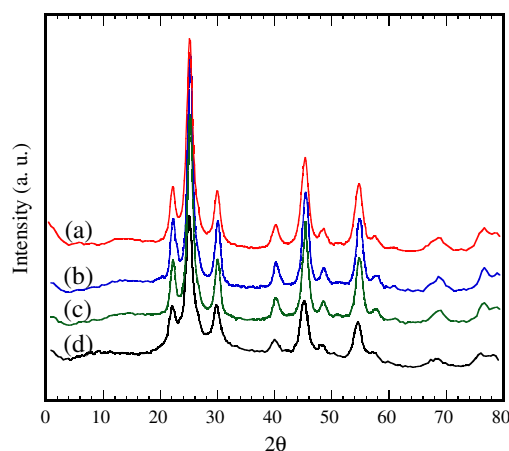


Fig. 1. XRD patterns of (a) ceria support, (b) oxidized, (c) reduced, and (d) sulfided $K_{0.05}Co_{0.6}20MoCe$.

The effects of the activation method on a $K_{0.05}Co_{0.6}20MoCe$ catalyst are shown in Fig. 2B. In Fig. 2B (g), three main peaks appeared on the sulfided catalysts. The first peak resulted from the reduction of a thiol group at 200–400 °C. The second peak resulted from the decomposition of the CoMoS phase at 700–900 °C ($Co_xMo_yS_z \rightarrow Co_9S_8 + MoS_2$), and the remaining peak resulted from the reduction of MoS₂ at 900–1000 °C [26]. There were three main peaks at 329, 831, and 993 °C for the sulfided samples. The three peaks were attributed to reduction of the -SH group, the decomposition of the so-called CoMoS phase, and reduction of MoS₂, respectively. In Fig. 2B (h), the same peaks as those for the sulfided samples were detected for presulfided plus reduced $K_{0.05}Co_{0.6}20MoCe$ except for a peak at 506 °C, but these peaks were smaller than those observed for sulfided samples, indicating that the reduction of presulfided catalysts naturally decreased the amount of active sites on the -SH group, the CoMoS phase, and the MoS₂ phase. The profile of the reduced catalyst had no specific peaks below approximately 400 °C, and broad peaks appeared above its temperature, as shown in Fig. 2B (i), indicating that the reduced catalyst was not reduced again below 400 °C, and the metal species resistant to reduction, such as bulk MoO₃, were reduced as the oxidized catalysts.

XPS measurements were performed on the samples that were pretreated by different methods. The binding energies and ratios of active metal species with different atomic valences are summarized in Tables 2 and 3. The XPS spectrum of Mo 3d was decomposed into five peaks by curve fitting based on Gaussian function: Mo⁴⁺ 3d_{5/2}, Mo⁴⁺ 3d_{3/2}, Mo⁵⁺ 3d, Mo⁶⁺ 3d_{5/2}, and Mo⁶⁺ 3d_{3/2}. As shown in Table 2, the doublet peaks of Mo⁴⁺ 3d for all samples showed two strong peaks for Mo⁴⁺ 3d_{5/2} and Mo⁴⁺ 3d_{3/2} at binding energies of 228.6 eV and 231.9 eV, respectively, which indicated the formation of MoS₂ species [27–30]. The weak doublet peaks at 232.6 eV and 235.7 eV were assigned to Mo⁶⁺ 3d_{5/2}, and Mo⁶⁺ 3d_{3/2}, respectively, ascribed to a part of the oxidic form that was still present after sulfidation. The peak for the Mo⁵⁺ species flattened and was subtle; it was characteristic of the oxysulfide MoO_xS_y phase with peaks at 230.0 eV [29,31]. The peak at 225.8 eV was assigned to S 2s [32]. The Co 2p spectrum contained at least three chemically shifted Co 2p_{3/2} peaks at 777.8 eV, 778.9 eV, 781.4 eV, which were assigned to the Co₉S₈ species, the Co–Mo–S phase, and Co²⁺ respectively, and the peak at approximately 785.0 eV was attributed to one shakeup peak [32,33]. Usually, the full sulfidation of Co²⁺ species is difficult and represents an interaction of Co with the support [32,34–36]. The shakeup peak for Co 2p was excluded from the composition of Co species. The S 2p spectrum was decomposed into two components at 161.5 eV and 162.8 eV, which were attributed to S²⁻ 2p and S₂²⁻ 2p, respectively. The binding energies were estimated with an accuracy of ±0.1 eV.

Table 3 shows that the S/Mo ratio, which represents the sulfidation degree, for the sulfided sample prior to reduction was smaller than that for the sample that was only sulfided. Simultaneously, the ratios of CoMoS/Co and Co₉S₈/Co for the presulfided + reduced sample were also lower than those for the sulfided sample. The degree of sulfidation for the presulfided + reduced sample was smaller than that for the sulfided sample. The ratio of Mo⁴⁺/Mo was in the following order: sulfided sample ≥ presulfided + reduced sample > reduced sample, and the ratio of (Mo⁵⁺ + Mo⁶⁺)/Mo was in the inverse order. When the sulfided sample was reduced, the ratio of Ce³⁺/Ce⁴⁺ was also increased. This finding indicates that the structure of the ceria support transformed from CeO₂ to Ce₂O₃.

3.2. CO hydrogenation

3.2.1. Ceria supported catalyst

To investigate the effects of the supports, C₂+ alcohol syntheses were performed over two catalysts supported on ceria and silica, i.e., $K_{0.05}Co_{0.6}10MoSi$ and $K_{0.05}Co_{0.6}10MoCe$, under the following conditions: 260 °C, H₂/CO ratio of 1.0, 5.0 MPa, and GHSV of 5000 h⁻¹. The results are shown in Table 4. The results from an Al-supported catalyst are

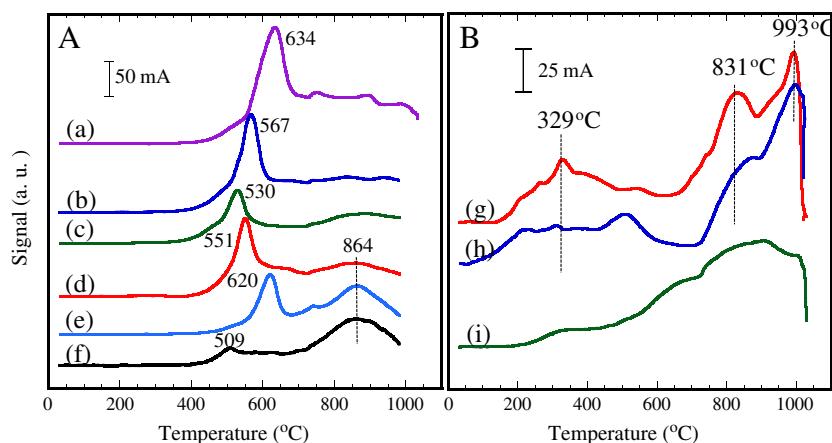


Fig. 2. H₂-TPR profiles of various oxidized catalysts: (a) oxidized K_{0.05}Co_{0.6}20MoCe, (b) oxidized K_{0.05}Co_{0.6}15MoCe, (c) oxidized K_{0.05}Co_{0.6}10MoCe, (d) oxidized K_{0.2}Co_{0.6}10MoCe, (e) oxidized K_{0.2}Co_{0.6}10MoSi, (f) oxidized K_{0.2}Co_{0.5}11MoAl, (g) sulfided K_{0.05}Co_{0.6}20MoCe, (h) presulfided + reduced K_{0.05}Co_{0.6}20MoCe, and (i) reduced K_{0.05}Co_{0.6}20MoCe.

also shown for comparison [7]. In CO hydrogenation, the main reaction products were methanol, C₂+ alcohols, C_{1–4} hydrocarbons, and CO₂. The formation of alcohols competed with F–T synthesis and a water–gas shift reaction during the CO hydrogenation reaction [2,37]. Among the three supports, CO conversion for the silica support was 0.9%, and the lowest. The selectivities for hydrocarbons, CO₂, and C₂+ alcohols were 37%, 30%, and 22%, respectively. With the ceria support, the selectivities for both hydrocarbons and CO₂ were suppressed to ca. 19% and the selectivity for C₂+ alcohols and CO conversion were 47% and 3.2%, respectively, which were higher than those for silica. Ceria showed the best performance of the three supports with regard to C₂+ alcohol synthesis.

3.2.2. Comparison of group VI metal–ceria catalysts

To compare the group VI metals as active metals on the ceria support, activity tests on three catalysts, K(C)_{0.05}Co_{0.6}MCo (M = 8.6Cr, 15Mo, 22W) at the constant M/Ce ratio of 0.23, were performed under the following conditions: 280 °C, H₂/CO ratio of 2.0, 5.0 MPa, and GHSV of 5000 h^{−1}. The CO conversion and selectivity for reaction products with catalysts based on different group VI metals are shown in Fig. 3. The space time yields (STYs) of C₂+ alcohols for the group VI metal-based catalysts are listed in Table 5. On the Cr-based catalyst, the CO conversion (8.9%) was approximately equal to that on the Mo-based catalyst and the order of conversion among the group VI metals was Cr = Mo > W. However, the selectivity for alcohols was the lowest for the Cr-based catalyst among the three catalysts. The selectivities for methanol and C₂+ alcohols were in the following order: Mo > W > Cr. Hydrocarbons were the dominant product on the

Cr-based catalyst, and the selectivity for hydrocarbons was 78%. The order of selectivity for hydrocarbons was in the reverse order of the selectivity for alcohols: Cr > W > Mo. There was little difference among the group VI metals with regard to the selectivity for CO₂, which was approximately 10%. Therefore, the selectivity for C₂+ alcohols, the CO conversion, and the STY of C₂+ alcohols on the Mo-based catalyst were higher than those on the W-based catalysts and reached 25%, 8.5%, and 0.61 mol/g-cat./h, respectively. The STY of C₂+ alcohols was in the following order, as shown in Table 5: Mo > W > Cr. Among the group VI metal-based catalysts, the Mo-based catalyst most favored the formation of C₂+ alcohols.

To investigate the stability in activity of the prepared catalysts, the activity test with sulfided K_{0.05}Co_{0.6}20MoCe catalyst was carried out under the condition of H₂/CO ratio of 2.0, 5.0 MPa, and 5000 h^{−1}. The reaction temperature was set at 260, 280, and 300 °C, and then returned at 260 °C again. The changes in the space–time yields of products with time on stream are shown in Fig. 4. At the first temperature of 260 °C, the space–time yields of methanol, C₂+ alcohols, hydrocarbons, and CO₂ were 0.23, 0.08, 0.21, and 0.03 mol/h/kg-cat., respectively. At the final temperature of 260 °C, those were 0.24, 0.07, 0.19, and 0.02 mol/h/kg-cat., respectively. It was observed that the activity for alcohol synthesis was almost constant, comparing at both temperatures of 260 °C. This means that no significant deactivation was observed during the period of at least 10.5 h.

3.2.3. Effects of Mo loading

The effects of Mo loading on catalytic activity were investigated on sulfided K_{0.05}Co_{0.6}MoCe catalysts at an H₂/CO ratio of 2.0, 5.0 MPa, and GHSV of 5000 h^{−1}, and the results are shown in Fig. 5. At 260 °C, when the Mo load increased from 11 to 15%, CO conversion and selectivity for C₂+ alcohols increased to 11% and 37%, respectively, and CO₂

Table 2

Binding energies of atomic components with various activated K_{0.05}Co_{0.6}20MoCe catalysts.

Binding energy (eV)		Activation method		
		Sulfided	Presulfided + reduced	Reduced
Mo 3d	Mo ⁴⁺ _{5/2}	228.70	228.64	228.60
	Mo ⁴⁺ _{3/2}	231.88	231.88	231.88
	Mo ⁵⁺ _{5/2}	230.11	229.93	230.11
	Mo ⁶⁺ _{5/2}	232.59	232.59	232.59
	Mo ⁶⁺ _{3/2}	235.79	235.79	235.64
	Co _{S8}	777.80	777.79	–
Co 2p	CoMoS	778.75	778.74	–
	Co ²⁺	781.40	781.30	781.14
	S 2s	225.80	225.71	225.62
S 2p	S ^{2−}	161.56	161.57	–
	S ^{2−}	162.81	162.79	–
	Ce ³⁺	886.15	886.15	886.17
Ce 3d	Ce ⁴⁺	881.94	881.91	881.90

Table 3

Ratios of atomic valence on various activated K_{0.05}Co_{0.6}20MoCe catalysts.

Atomic ratio (–)	Activation method		
	Sulfided	Presulfided + reduced	Reduced
S/Mo	0.86	0.72	–
CoMoS/Co	0.40	0.34	–
Co _{S8} /Co	0.10	0.063	–
Co ²⁺ /Co	0.50	0.25	0.28
CoMoS/Co _{S8}	4.2	5.3	–
Mo ⁴⁺ /Mo	0.65	0.66	0.41
(Mo ⁵⁺ + Mo ⁶⁺)/Mo	0.35	0.34	0.59
Ce ³⁺ /Ce ⁴⁺	1.7	2.2	2.5
K/Mo	0.18	0.11	0.18
Co/Mo	0.34	0.17	0.07

Table 4CO conversion and selectivity for sulfided and supported catalysts at 260 °C, H₂/CO = 1.0, 5.0 MPa, and 5000 h⁻¹.

Catalyst		K ₀₂ Co ₀₆ 10MoCe	K ₀₂ Co ₀₆ 10MoSi	K(C) ₀₂ Co ₀₅ 11MoAl [9]
CO conversion (%)		3.2	0.9	5.9
Selectivity (%)	Methanol	15	11	3.6
	C ₂₊ alcohols	47	22	17
	Ethanol	41	20	16
	2-Propanol	0.9	0.5	0.2
	<i>n</i> -Propanol	4.0	1.6	0.6
	2-Butanol	0.3	0.0	0.4
	<i>n</i> -Butanol	1.0	0.3	0.0
	Hydrocarbons	19	37	46
	CH ₄	11	32	25
	C ₂ H ₆	4.0	5.3	16
	C ₃ H ₈	3.0	0.0	4.3
	<i>n</i> -C ₄ H ₁₀	1.4	0.0	0.8
	CO ₂	19	30	33

selectivity also increased slightly, while the selectivities for methanol and hydrocarbons were decreased. At more than 15% Mo loadings, C₂₊ alcohol selectivity became almost constant at 38% whereas the CO conversion was dropped to 7.7%. The optimum Mo load was 15%. The conversion and all selectivities at 280 °C resembled those at 260 °C in the curve of the graph. At 300 °C, the CO conversion and the selectivities for methanol, C₂₊ alcohols, and CO₂ were similar to those at the other temperatures. The gradient for the selectivity curve for hydrocarbons decreased with temperature increased from 260 °C to 280 °C, and thus, at the even higher temperature of 300 °C, the formation of hydrocarbons was more favorable. Therefore, at 300 °C, this trend was different from the other temperatures.

3.2.4. Effect of activation methods

The effects of the activation methods on a K₀₀₅Co₀₆20MoCe catalyst were investigated under the following conditions: H₂/CO ratio of 2.0, 5.0 MPa, and GHSV of 5000 h⁻¹. The results are shown in Fig. 6. The ordering trends for conversion and for almost all the selectivities were similar to the respective temperatures within a range of 260–300 °C. Of the three activating methods, the CO conversion and selectivity for C₂₊ alcohols were the highest and the selectivity for hydrocarbons was the lowest for the sulfided sample. In contrast, the CO conversion and selectivity for C₂₊ alcohols were the lowest and the selectivity for hydrocarbons was the highest for the reduced samples. The selectivities for C₂₊ alcohols and hydrocarbons were moderate for the presulfided and reduced samples.

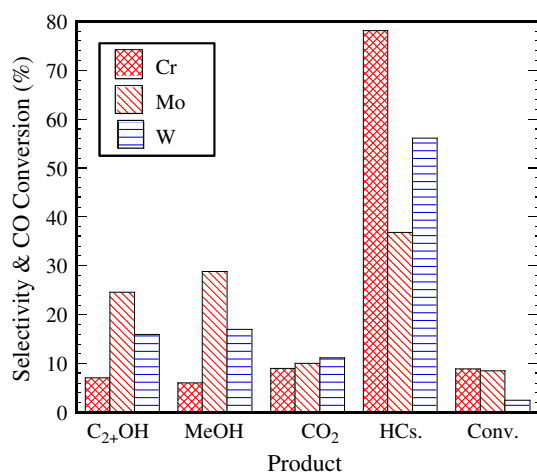


Fig. 3. CO conversion and selectivity for the synthesis of C₂₊ alcohols on presulfided + reduced K(C)₀₀₅Co₀₆MCE (M = Cr, Mo, W) at 280 °C, H₂/CO = 2.0, 5.0 MPa, and 5000 h⁻¹. a) HCs: hydrocarbons, b) Conv.: CO conversion.

Among the three activation methods, the selectivity for C₂₊ alcohols and CO conversion decreased in the following order: sulfided > presulfided + reduced > reduced, whereas the selectivity for hydrocarbons decreased in the reverse order. The methanol selectivity decreased in the following order: reduce = presulfided + reduced > sulfided catalyst, except for at 260 °C. There was little difference among the activation methods with regard to the selectivity for methanol at 260 °C. The CO₂ selectivity decreased in the following order: sulfided > reduced ≥ presulfided + reduced catalyst. Therefore, the formation of C₂₊ alcohols over the sulfided catalyst was more favorable than that over the reduced or presulfided + reduced catalyst.

4. Discussion

Among the silica, ceria, and alumina supports, ceria supports showed the most favorable results, and they suppressed the formation of CO₂ (Table 5). It has been suggested that ceria can suppress CO₂ through storing a large amount of oxygen (high oxygen-storage capacity (OSC)). Ceria can also serve as a support for catalysts in exhaust-gas treatment and steam reforming reactions, and it is known as a strong reducing agent [15,16]. A mechanism of adsorbed CO species on active sites on a ceria-supported catalyst is shown in Scheme 1. It has been proposed that CO₂ is formed via the reaction of the non-dissociatively adsorbed CO* species (or CO into feed gas) with O* species derived from dissociatively adsorbed C–O species [7,18,38]. However, the ceria support could accept and store oxygen molecules or dissociated oxygen species derived from dissociative C–O adsorption via transformation from the Ce₂O₃ (Ce³⁺) phase to the CeO₂ (Ce⁴⁺) phase. Thus, it is possible that the ceria support suppressed the formation of CO₂ in CO hydrogenation, as shown in Scheme 1. This idea is consistent with the fact that the ratio of Ce³⁺/Ce⁴⁺ decreased when the sulfided sample was reduced, as shown in Table 3, indicating that the ceria readily transformed from CeO₂ to Ce₂O₃. Ce₂O₃ was recovered to CeO₂ through the reaction of easily and dissociatively adsorbed H* species with Ce₂O₃.

Among group VI metals, the Mo-based catalyst favored the formation of C₂₊ alcohols, as noted in Section 3.2.1, and the order of STY for C₂₊ alcohols was Mo > W > Cr. The mixed metal sulfide phase, such as the so-called Co–Mo–S phase, was reported to be formed over sulfided Co–Mo catalysts, and it significantly improved the catalytic

Table 5Effects of group VI metals on space time yield of alcohols over presulfided plus reduced K(C)₀₀₅Co₀₆MCE (M = Cr, Mo, W) at 280 °C, H₂/CO = 2.0, 5.0 MPa, and 5000 h⁻¹.

Catalyst	STY (mol/h/kg-cat.)	
	C ₂₊ alcohols	Methanol
K(C) ₀₀₅ Co ₀₆ 8.6CrCe	0.074	0.15
K(C) ₀₀₅ Co ₀₆ 15MoCe	0.61	1.5
K(C) ₀₀₅ Co ₀₆ 22WCE	0.28	0.62

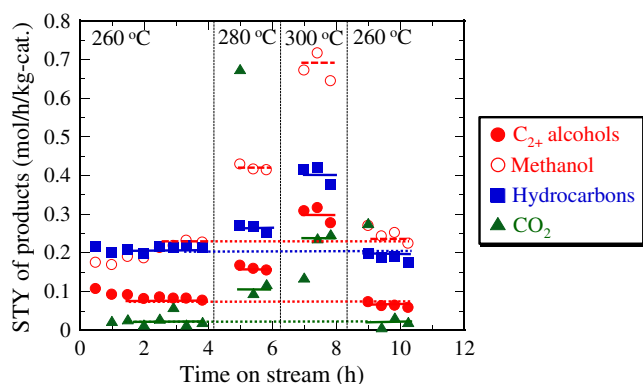


Fig. 4. Changes in space-time yields of products with time on stream under condition of $K_{0.05}Co_{0.620}MoCe$, H_2/CO ratio of 2.0, 5.0 MPa, and 5000 h^{-1} .

performances for hydrodesulfurization and hydrodenitrogenation reactions [12]. The Cr-based catalyst resists forming mixed metal sulfide phases, namely Co–Mo–S and Co–W–S phases, compared to the Mo or W-based catalysts [39]. The sulfidation of the W-based catalysts is more difficult than that for the Mo-based catalysts because the tungsten–sulfur bond strength was slightly weaker than the molybdenum–sulfur bond strength [12,40]. It was also reported that a considerable amount of W was present in the oxidic form after sulfidation at 673 K [41,42]. In that case, the formation of active sites on mixed-metal phases such as the Co–W–S phase on Co-doped WS_2 catalysts was more difficult than that on Co-doped MoS_2 catalysts. Therefore, the easiness of forming mixed-metal sulfide phases among group VI metals decreased in the following order: Mo > W > Cr. This sequence was identical to that of the selectivity for C_{2+} alcohols. The Co–Mo–S or Co–W–S phase played an important role in the formation of C_{2+} alcohols.

In the present study, as shown in Table 4, the reaction products of the Mo-based catalysts prepared were mainly composed of linear hydrocarbons and linear C_{1-4} alcohols. However, a small amount of non-linear alcohols such as 2-propanol and 2-butanol were also produced. Fujimoto et al. reported that the reaction products using the MoS_2 -based catalysts, which were classified into the catalysts for F–T synthesis, were mainly composed of linear C_{1-4} alcohols. They proposed the synthesis in branched alcohols, i.e., alcohol condensation reaction, as well as the synthesis in linear alcohols, may occur over the Mo-based catalysts simultaneously [43,44]. Thereby, alcohol condensation as well as F–T synthesis may also happen during the alcohol synthesis over the Mo-based catalysts prepared in this study.

While the optimum loading amount of Mo was 15%, as shown in Fig. 5, CO conversion was depressed at a Mo load of 20%. Increases in the loading amount of metal generally lead to a large number of active sites. However, the BET surface area decreased from $73\text{ m}^2/\text{g}$ to $48\text{ m}^2/\text{g}$, when the Mo load was increased from 15% to 20%, as shown in Table 1. It was thought that the dispersion and number of active sites of supported metals on the catalyst would be lower at a 20% load than at a 15% load. As also mentioned for the H_2 -TPR analyses, the reduction peak for Mo^{6+} to Mo^{4+} species was shifted to high temperature with increasing Mo load, as shown in Fig. 2, indicating the formation of a bulk MoO_3 phase that was difficult to reduce on the catalysts.

Regarding the activation methods, the sulfided $K_{0.05}Co_{0.620}MoCe$ catalyst showed superior performance for the synthesis of C_{2+} alcohols to that of the presulfided + reduced catalyst. The H_2 -TPR measurements on the presulfided + reduced catalyst indicated that the amounts of thiol groups, the MoS phase, and the CoMoS phases were decreased, compared to the sulfided catalyst. Moreover, as noted above, XPS analysis showed that the outer CoMoS phase on the sulfided catalyst was larger than that on the presulfided + reduced catalyst. Consequently, the thiol groups, the MoS phase, and the CoMoS phase would produce a catalyst selective for C_{2+} alcohols.

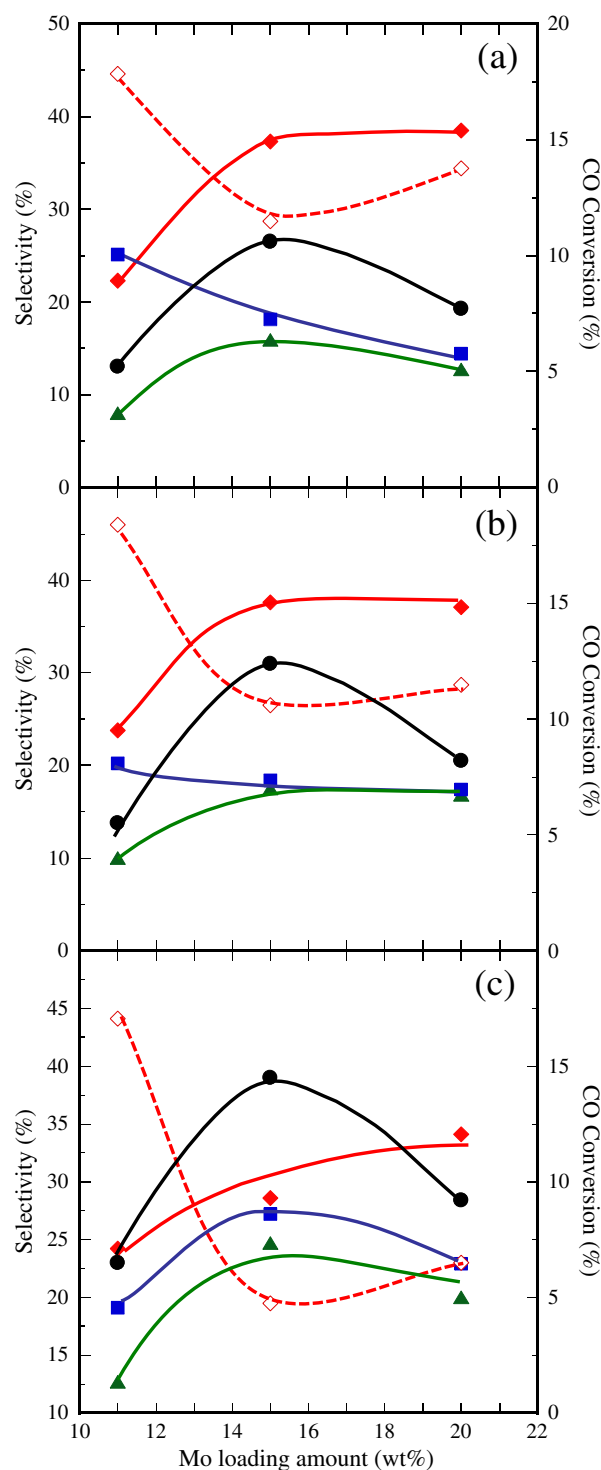


Fig. 5. Effects of Mo loading amount on CO conversion and selectivities for $K_{0.05}Co_{0.6}MoCe$ at (a) 260 °C, (b) 280 °C, (c) 300 °C, $H_2/CO = 2.0$, 5.0 MPa, 5000 h^{-1} (● CO Conversion, ♦ C_{2+} alcohols, ◊ Methanol, ■ Hydrocarbons, ▲ CO_2).

5. Conclusions

The effects of ceria, group VI metals, loading amount of Mo, and activation methods on C_{2+} alcohol synthesis were investigated in this study. The selectivity for C_{2+} alcohols and the CO conversion with a ceria-supported catalyst were 47% and 3.2%, respectively. Ceria supports showed the best performance among silica, ceria, and alumina supports. The formation of CO_2 was suppressed using ceria, which is capable of storing oxygen species. Among the group VI metals investigated, the

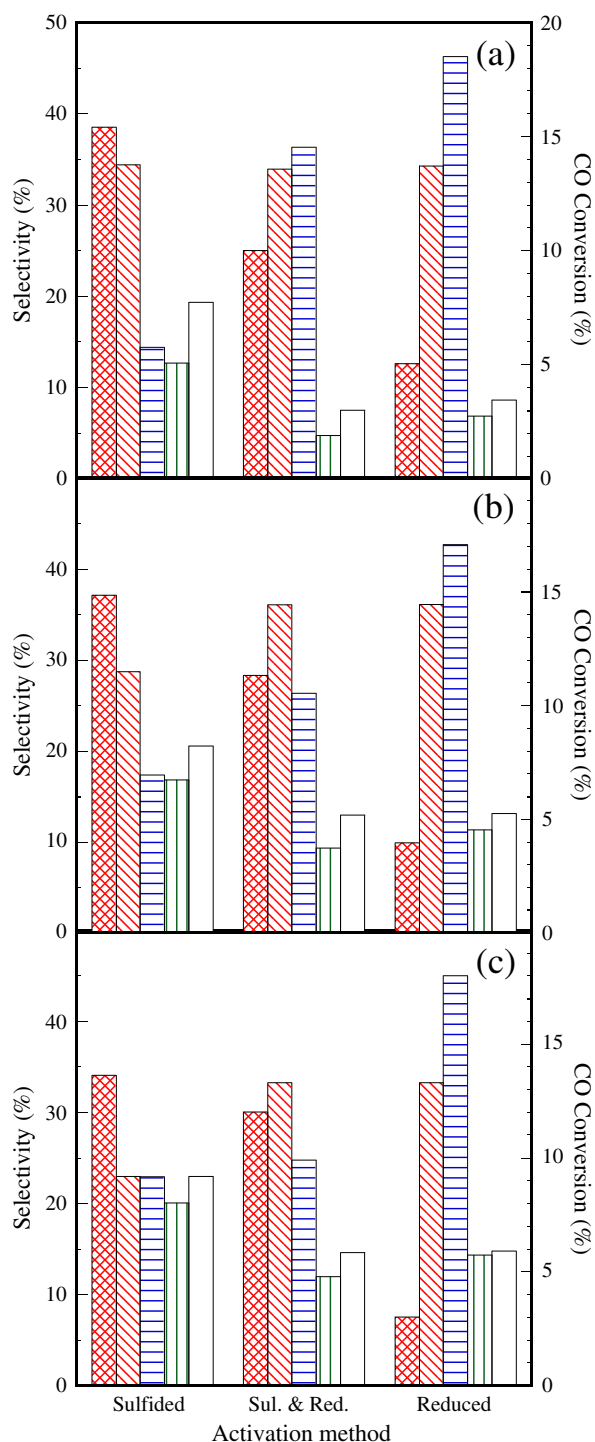
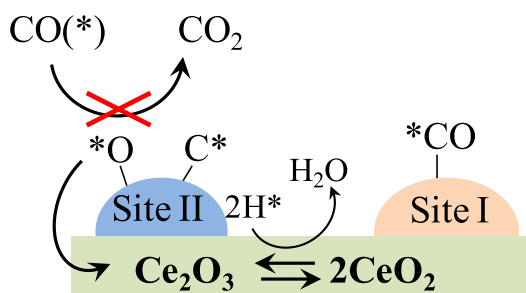


Fig. 6. Effects of activation methods on CO conversion and selectivities for $K_{0.05}Co_{0.620}MoCe$ catalyst at (a) 260 °C, (b) 280 °C, (c) 300 °C, H_2/CO ratio of 2.0, 5.0 MPa, and GHSV of $5000\ h^{-1}$. $\blacklozenge$: C_2+ alcohols, $\color{red}\lozenge$: Methanol, $\blacksquare$: Hydrocarbons, $\blacktriangle$: CO_2 , $\square$: Conversion.

Mo-based catalyst favored the formation of C_2+ alcohols, and the order of STY for C_2+ alcohols was $Mo > W > Cr$. This trend was consistent with the easiness of forming mixed-metal sulfide phases among group VI metals. Moreover, for group VI metals, the Co–Mo–S phase played an important role in the formation of C_2+ alcohols. With Mo loads ranging from 11 to 20%, at Mo loads of more than 15%, C_2+ alcohol selectivity stayed almost constant at 38%, whereas the CO conversion at Mo loads of 20% dropped to 7.7% at 260 °C. Decreasing BET surface area and the formation of bulk MoO_3 on the catalysts indicated that dispersion



Site I: non-dissociative CO adsorption

Site II: dissociative CO adsorption

Scheme 1. Suppression of CO_2 by oxygen storage capacity of ceria support in CO hydrogenation.

and sites of the active metals were decreased. Regarding the activation methods, the sulfided $K_{0.05}Co_{0.620}MoCe$ catalyst favored the selectivity for C_2+ alcohols. H_2 -TPR and XPS analyses showed that the MoS phase, thiol groups, and the Co–Mo–S phase would enhance catalyst selectivity for C_2+ alcohols.

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